FixKar

Project Proposal

Supervisor
Mr. Usama Khalid

Submitted by

Mr. Muhammad Osama
2280151

Ms. Zainab Asif
2280134

Faculty of Computing and Engineering Sciences
Shaheed Zulfiqar Ali Bhutto Institute of Science and Technology, Karachi.

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1. Introduction

In emergency and routine vehicle breakdown situations, timely access to a skilled mechanic is often a challenge, especially when stranded or unaware of nearby services. FixKar is a mobile-based solution that connects users to nearby verified mechanics within a 1 km radius in emergencies and allows scheduled services for non-emergency cases. The app bridges the gap between users and reliable roadside assistance, bringing real-time help to fingertips with transparency, affordability, and AI support..

2. Objective

To design and develop a real-time mobile application that connects users with nearby verified mechanics for emergency and scheduled services, incorporating AI-based vehicle assistance, featuring live tracking, user reviews, and admin-level monitoring through a web dashboard.

3. Problem Description

Vehicle breakdowns, whether on highways or within cities, often leave users helpless, especially during off-hours or in unfamiliar areas. Traditional towing or workshop discovery is slow, non-transparent, and lacks verified credibility of service providers. Additionally, routine vehicle care such as oil changes or general inspections lack easy booking options with at-home availability.

FixKar addresses:

- The absence of a verified, transparent, and real-time mechanic booking system.
- Lack of live location tracking during mechanic dispatch, causing uncertainty for users.
- Emergency roadside situations requiring nearby immediate help.
- No feedback loop or rating system to filter poor service quality.
- Communication and education gaps, especially for less-tech-savvy mechanics.
- Absence of intelligent AI guidance for vehicle-related queries.

This system brings affordability, accountability, and informed decision-making to the mechanic-user interaction through reviews, live tracking, geolocation, chat, and verified onboarding.

4. Target Industry

The target industry is the automobile assistance and emergency response sectors.

5. Methodology

- The mobile and web application will be built using MERN Stack.
- Google Maps API will be used for real-time location tracking.

- WebSockets (Socket.IO) will enable real-time chat between users and mechanics and handle live booking status updates.
- Mechanic identity verification will be conducted via the Tasdeeq using CNIC information (manually).
- Payments will be processed using Stripe API, and the app will retain 10% commission per successful booking.
- AI-based vehicle support will be provided through OpenRouter (GPT), limited to vehicle-related queries.
- Notifications will be handled using FCM.
- Image uploads will be handled using Cloudinary.

6. Project Scope

FixKar will focus on connecting vehicle users with verified mechanics, including:

- Mechanic registration and verification by admin via CNIC using Tasdeeq Verification manually.
- Emergency bookings based on real-time location, with options for upfront or post-inspection pricing.
- Live GPS tracking of mechanics during active bookings.
- Non-emergency services via scheduled appointments.
- In-app chat between users and mechanics for better coordination.
- AI-based vehicle assistance using GPT (vehicle-related queries only).
- Multilingual UI (English & Urdu) and video tutorials for mechanics (opt).
- Ratings and reviews from users after service completion.
- Transparent billing, viewable by both user and mechanic.
- Mechanic pays 10% commission to the app via Stripe after successful service completion.
- Real-time notifications for clients, mechanics will be handled through the app across all modes.
- Admin oversight of users, mechanics, bookings, and revenue analytics.

For this project, the system will not cover multi-mechanic or collaborative bookings, advanced vehicle diagnostics beyond AI text assistance, offline app functionality, integrations with external services (such as insurance providers or towing companies). These features may be considered for future enhancements.

7. Feasibility Study

With above defined scope, we are hopeful that we'll complete the project within the time constraints defined.

Risks Involved:

- **Technical literacy gap:** Mitigated through in-app video tutorials and a bilingual interface (English & Urdu).

- **Fake or unverified mechanic registrations**: Addressed by using Tasdeeq API manually for CNIC-based verification, ensuring only credible mechanics are allowed on the platform.
- **AI misuse**: Controlled by limiting GPT-based responses strictly to vehicle-related queries using system-level instructions.
- **Mechanic non-payment of commission**: Managed by setting payment deadlines and automatic fines. Repeat violations may result in account suspension or permanent blacklisting.
- **User refusal after mechanic's inspection (post-inspection pricing model):** If a mechanic quotes the price after inspection and the user declines service, a standard Rs 250 inspection visit fee is enforced to compensate the mechanic's time and travel.
- **Mechanic no-show or overcharging**: Managed via review system, shared bill transparency, and fixed arrival charges if the service is declined post-inspection.

Resource Requirement:

- Development laptop with internet access
- Free-tier access to APIs (Google Maps, Stripe, Tasdeeq, OpenRouter)
- WebSocket hosting (using Socket.IO) for real-time communication.
- FCM for notifications.
- Cloud-based database and storage: MongoDB Atlas (data) and Cloudinary (media uploads).

8. Solution Application Areas

This solution has high real-world value in the automobile assistance and emergency response sectors. It specifically targets:

- Roadside Assistance Industry
- Urban/Remote Mobility
- Gig Economy for Mechanics

Benefits:

- Reduces waiting time during emergencies
- Improves service reliability through identity verification and ratings
- Creates income opportunities for local service providers

9. Tools/Technology

Frontend:

- React Native (User and Mechanic Apps)
- ReactJS (Admin Panel)
- Bootstrap (UI Styling)

Backend:

- NodeJS (API development)
- MongoDB (Database management)
- Socket.IO (Real time Chats)

APIs/Integrations:

- Google Maps (Location services)
- OpenRouter GPT API (Restricted chatbot)
- Stripe API (Payments)

Other Tools:

- GitHub
- Postman
- Cloudinary (for hosting videos and images)
- Firebase Services (FCM)
- AWS and Vercel (Hosting App and Web)

10. Expertise of the Team Members

Both team members are well-equipped with the skills required for successful project execution. We have strong command over the MERN stack, real-time communication using WebSocket's, and API integrations. We are adaptive and quick learners, ready to explore and integrate any new technology required during development. This project aligns with our technical strengths and is of equal interest to both team members

11. Milestones

FYP-1:

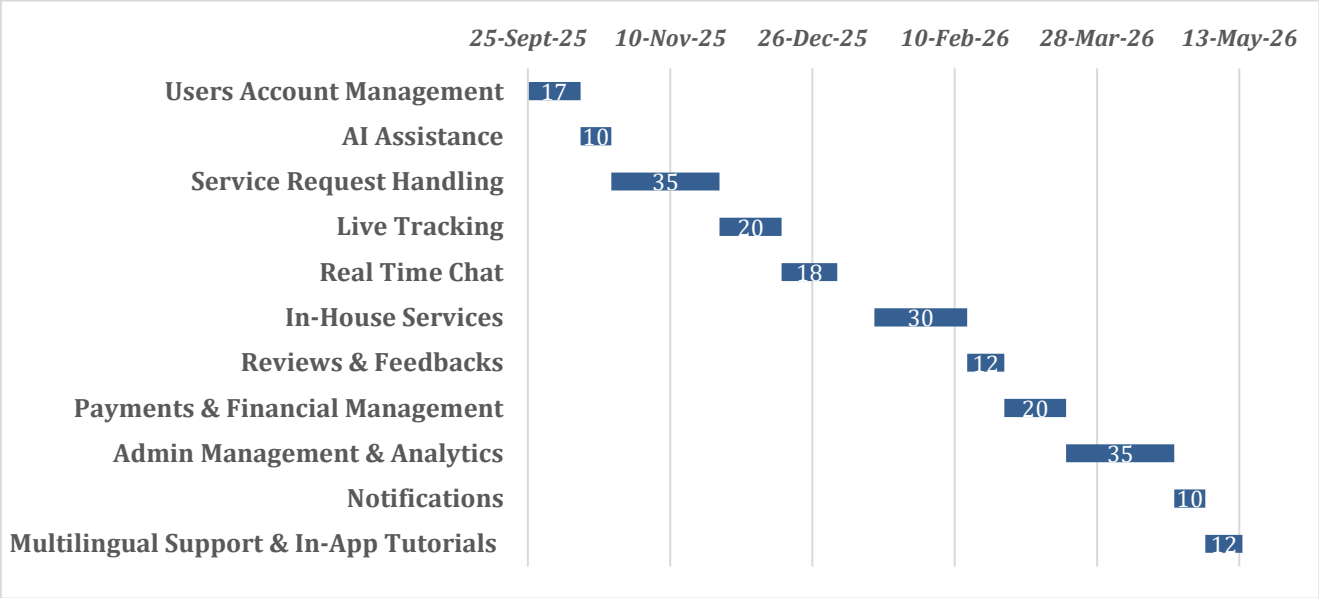
- Users shall be able to sign up, log in, recover their password using the "forgot password" feature, and update their profiles. Mechanic accounts shall require admin approval via Tasdeeq verification(manually) before activation.
- Clients shall be able to request mechanics for assistance; nearby available mechanics can respond to these requests in real time and either offer an instant quote or provide a price after inspection.
- Clients shall be able to track the live location of the assigned mechanic on a map.
- Clients and mechanics shall be able to communicate via real-time chat.
- Clients shall have access to an AI assistant that responds only to vehicle-related queries.

FYP-2:

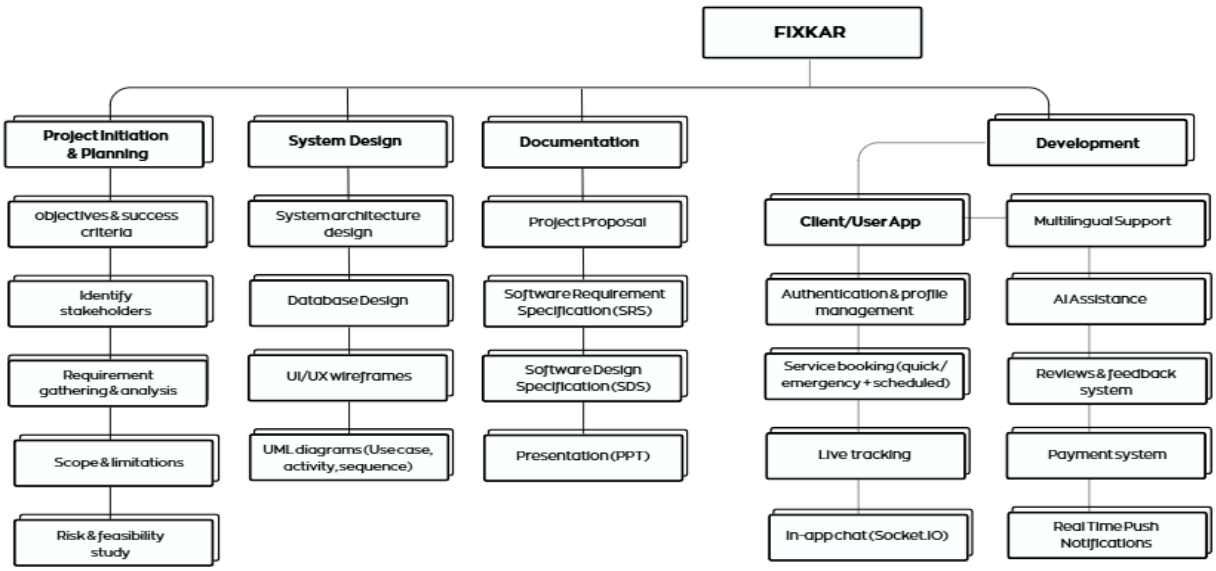
- Clients shall be able to book Fixkar's in-house services with doorstep appointment scheduling.
- Clients shall be able to submit feedback and reviews once the service is completed.

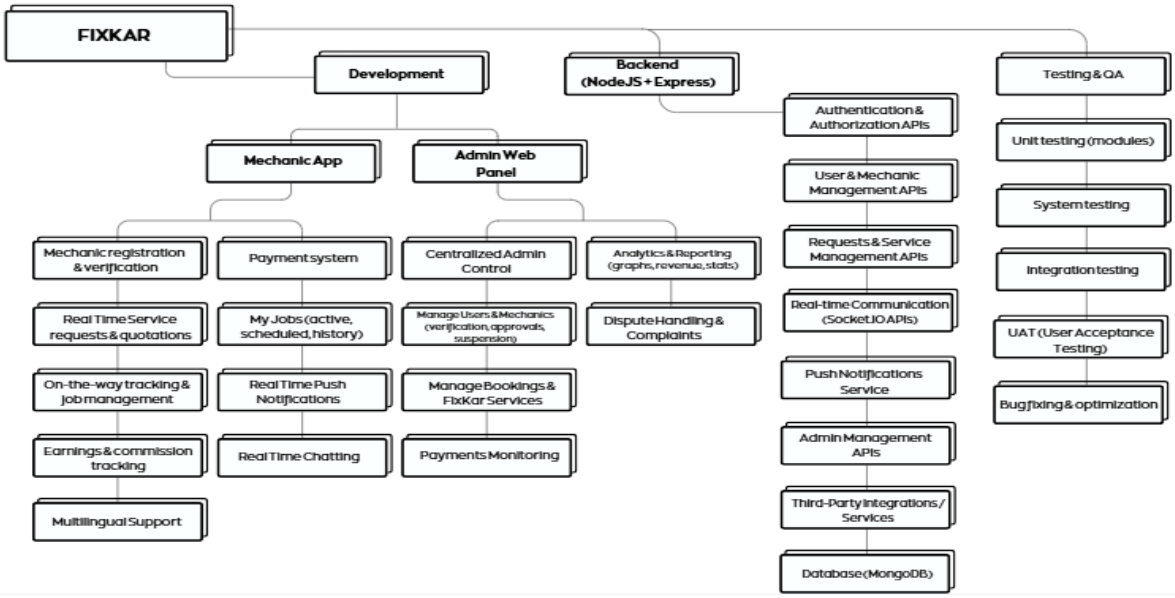
- Admin shall be able to manage users, mechanics, bookings, and financial operations, including payments and commission distribution, while also monitoring system analytics.
- Clients shall be able to pay for services through an integrated payment system; both admin and mechanic shall be able to manage and split payments via the system.
- Real-time notifications for clients, mechanics will be handled through the app across all modes.
- The mechanic app shall offer multilingual support and include in-app tutorial videos for ease of use. (optional)

12. Project Schedule



13. Work Breakdown Structure





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